



Medical Microbiology

Lecture Eight Biochemical Test

Third stage- College of Dentistry

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Introduction:

- Biochemical tests are the tests used for the identification of bacteria species based on the differences in the biochemical activities of different bacteria.
- These differences include carbohydrate metabolism, protein metabolism, fat metabolism, production of certain enzymes, ability to utilize a particular compound, etc.

Biochemical tests include:

1. Mannitol Salt Agar (MSA):

- MSA is both a **selective** and **differential** medium. Its high salt concentration allows the growth of salt-tolerant bacteria, particularly **Staphylococcus** species.
- The **differential component** is **mannitol**, which some bacteria can ferment, producing acidic by-products that lower the PH.
- The drop in pH causes the **phenol red** indicator in the medium to turn **yellow**.
- **Staphylococcus aureus** ferments mannitol and produces a yellow halo, while **Staphylococcus epidermidis** does not ferment mannitol and therefore does **not** cause a color change.



2. MacConkey Agar:

- MacConkey agar is widely used to distinguish members of the **Enterobacteriaceae** family.
- MacConkey agar is a **selective** and **differential** medium.
- The selective agents, **bile salts** and **crystal violet**, inhibit **Gram-positive bacteria**, allowing only Gram-negative organisms to grow.
- The **differential component** is **lactose**. When bacteria ferment lactose, they produce acid, lowering the pH.
- The pH indicator **neutral red** turns **bright pink** in acidic conditions, so lactose-fermenting bacteria such as *Escherichia coli* form **pink colonies**.



Lactose non fermentor

Lactose Fermentor

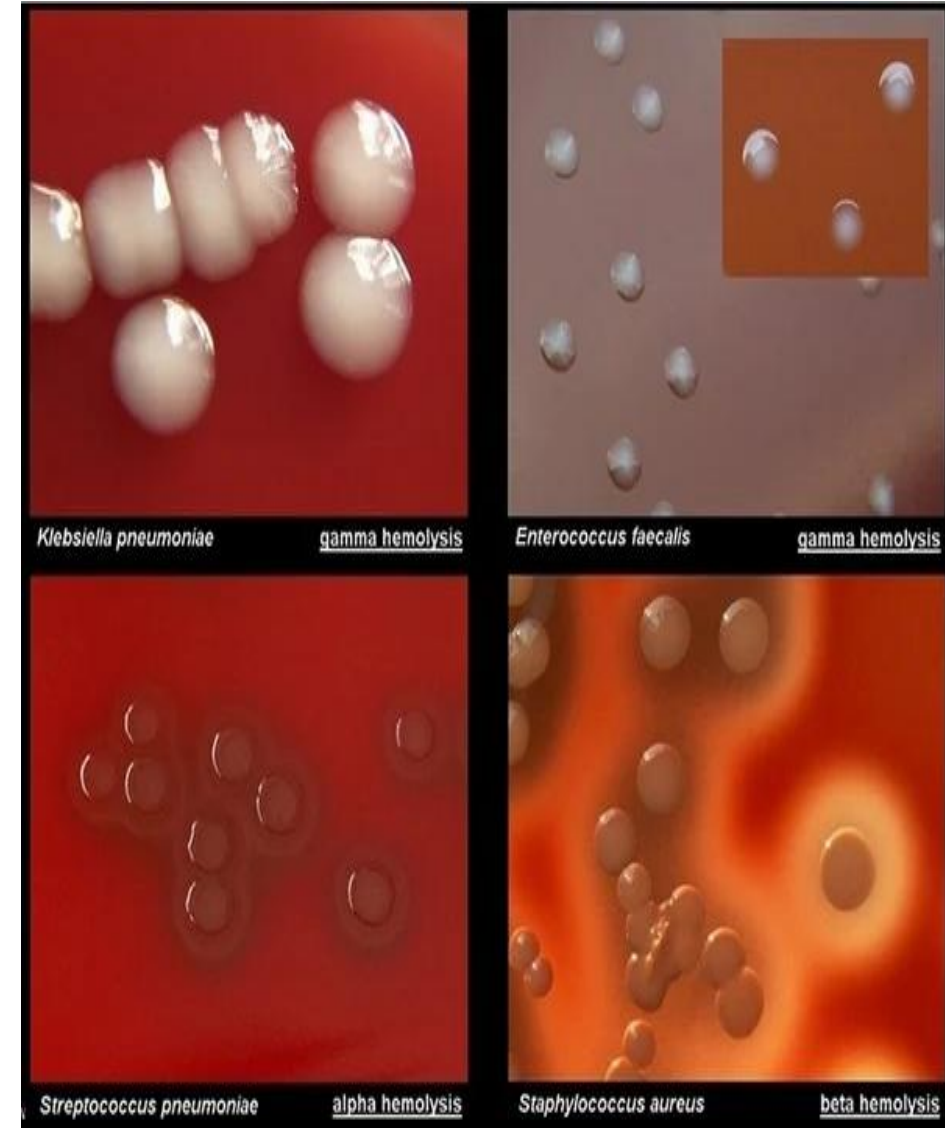
3. Blood Agar Plates (BAP):

- Blood agar is a **differential** medium composed of a nutrient-rich base supplemented with **5% sheep red blood cells**.
- It is used to assess whether bacteria produce **hemolysin**, (enzyme that damage or lyse red blood cells).
- The **type of hemolysis** produced helps differentiate bacterial species:

1-Beta-hemolysis: *Complete* hemolysis, seen as a **clear, transparent zone** around the colonies (e.g., *Streptococcus pyogenes*).

2-Alpha-hemolysis: *Partial* hemolysis, producing a **greenish, opaque zone** around the colonies (e.g., *E. coli*).

3-Gamma-hemolysis: *No* hemolysis, with **no visible change** in the agar surrounding colonies (e.g., *Staphylococcus epidermidis*).



4-Methyl Red (MR) Test:

This test determines whether a bacterium can ferment glucose and produce **strong, stable acids** as end products. The organism is first grown in a glucose-containing broth.

After incubation, **methyl red indicator** is added. If the bacteria perform mixed-acid fermentation, it lowers the pH enough to turn the methyl red indicator **from yellow to a bright red**, indicating a **positive result**. Common **MR-positive bacteria** include **Escherichia coli** and **Salmonella spp.**



5–Voges Proskauer (VP) Test:

The VP test detects the ability of bacteria to convert **glucose** fermentation products into **acetoin** (a neutral end product). After incubation in a glucose phosphate broth, alpha-naphthol and potassium hydroxide (KOH) are added.

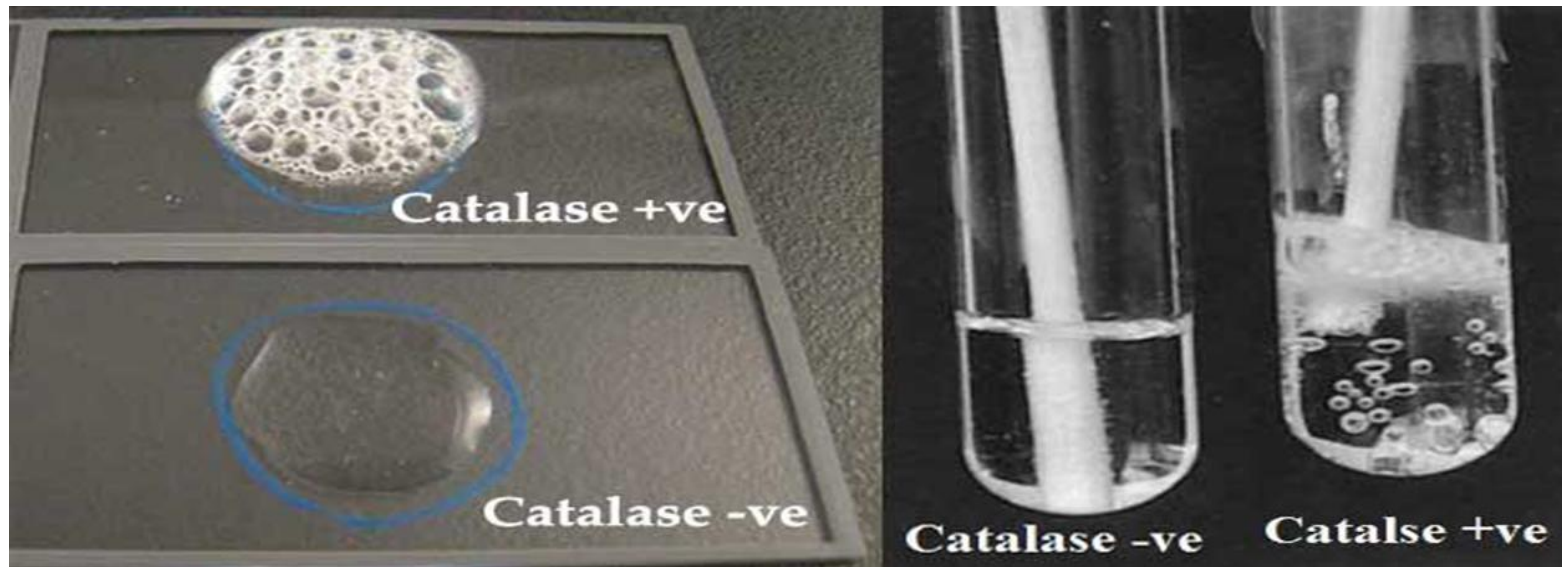
If acetoin is present, the reagents oxidize it to diacetyl, producing a characteristic red color, indicating a positive VP reaction.

Common VP-positive organisms include **Enterobacter** and **Klebsiella species**.



6- Catalase Test:

- This test is used to identify organisms that produce catalase enzyme.
- This enzyme detoxifies hydrogen peroxide by breaking it down into water and oxygen gas.
- The bubbles resulting from production of oxygen gas clearly indicate a catalase positive result.
- e.g./Staphylococcus spp. (+ve), Streptococcus spp. (–ve) .

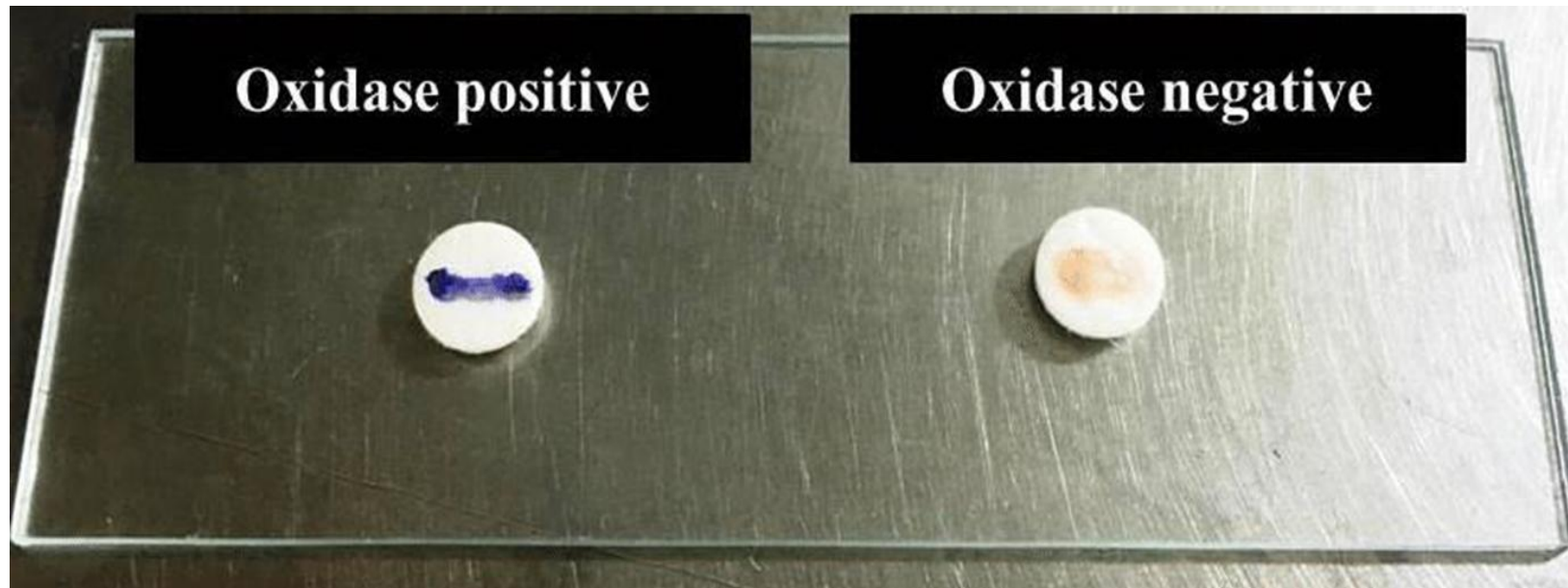


7- Oxidase Test :

The oxidase test is a simple biochemical test used to determine whether a microorganism produces the enzyme **cytochrome c oxidase**, which plays a vital role in the electron transport chain during aerobic respiration.

When a colony is exposed to an oxidase reagent bacteria that contain cytochrome oxidase will rapidly oxidize the reagent, producing a **dark purple** or **blue** color within seconds. This color change indicates a **positive** oxidase reaction.

Common oxidase-positive bacteria include **Pseudomonas**, **Neisseria**, and **Vibrio** species.

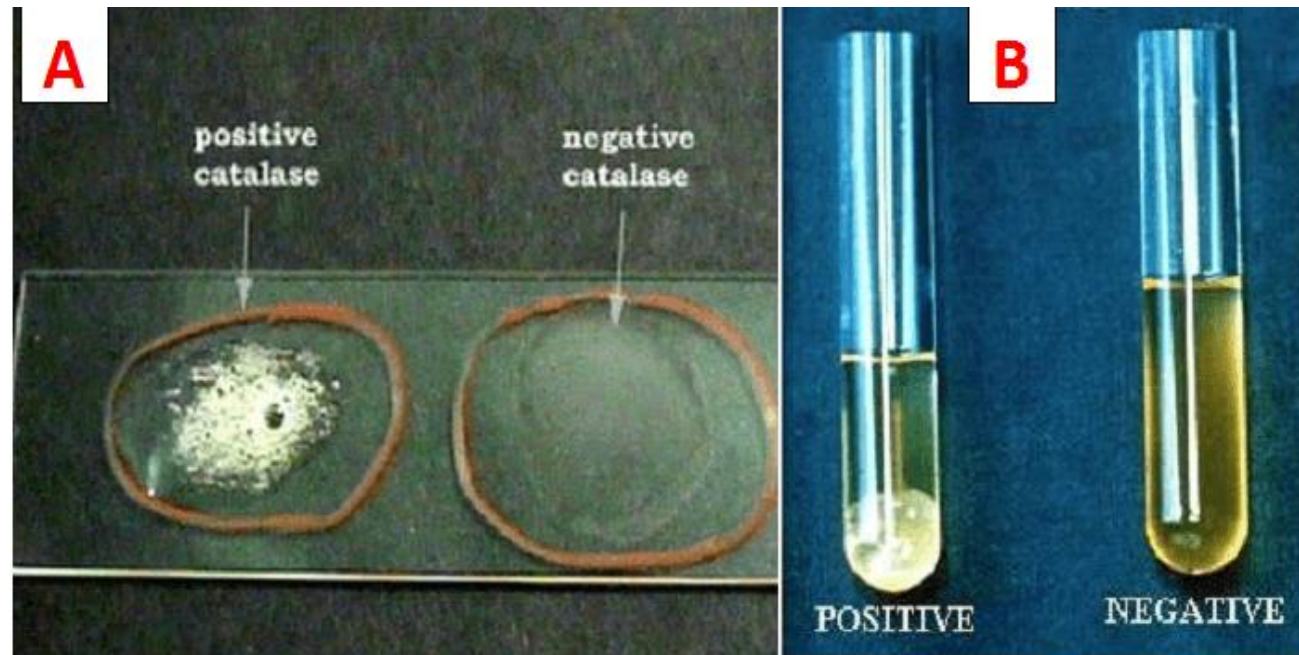


8–Coagulase Test:

It is the test that detects the enzyme **coagulase**, which has the ability to clot blood plasma. It is primarily used to identify **coagulase-positive *Staphylococcus aureus***.

Coagulase is an important **virulence factor** of *S. aureus*. By forming a fibrin clot around the bacteria, it helps shield the organism from **phagocytosis**, enhancing its survival in host tissues.

This test is essential for distinguishing *S. aureus* from other **coagulase-negative staphylococci (CoNS)**.

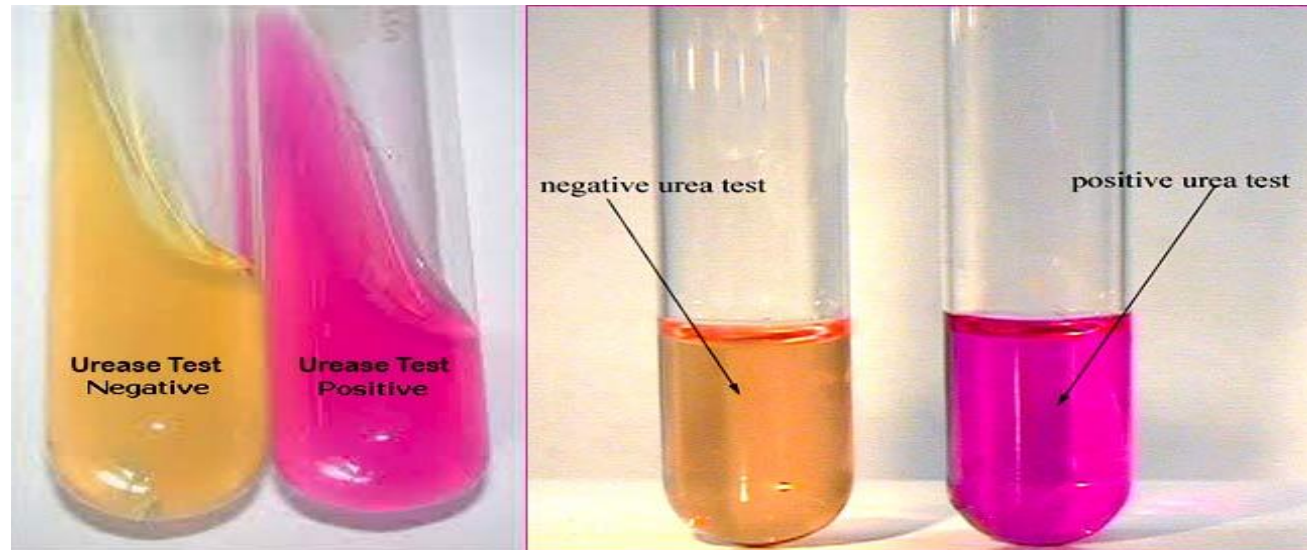


9–Urease Test:

The urease test identifies bacteria that produce the enzyme **urease**, which hydrolyzes urea into **ammonia** and **carbon dioxide**.

It is especially useful for differentiating **Proteus species**, which are strongly urease-positive, from other enteric bacteria.

As urease breaks down urea, the released **ammonia** increases the pH of the medium above **(8.4)**. This alkaline shift causes the pH indicator **phenol red** to change from **yellow** to a **bright pink**, indicating a **positive urease reaction**.



10–Indole Test :

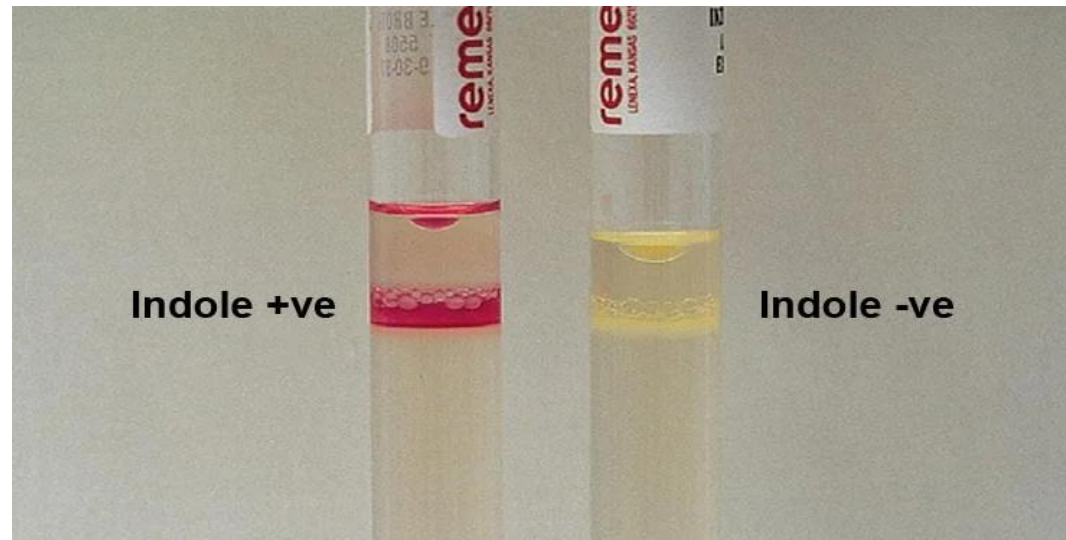
The indole test is a biochemical test used to determine whether a bacterium can degrade the **amino acid tryptophan** to produce **indole**, using the enzyme **tryptophanase**.

The organism is first grown in a tryptophan-rich medium, after incubation, **Kovac's reagent** is added.

If indole is present, the reagent reacts with it to form a red layer on the surface—this indicates a **positive** indole test.

No color change or a yellow layer indicates a negative result.

The test is routinely used to help identify indole-positive bacteria, most notably **Escherichia coli**.

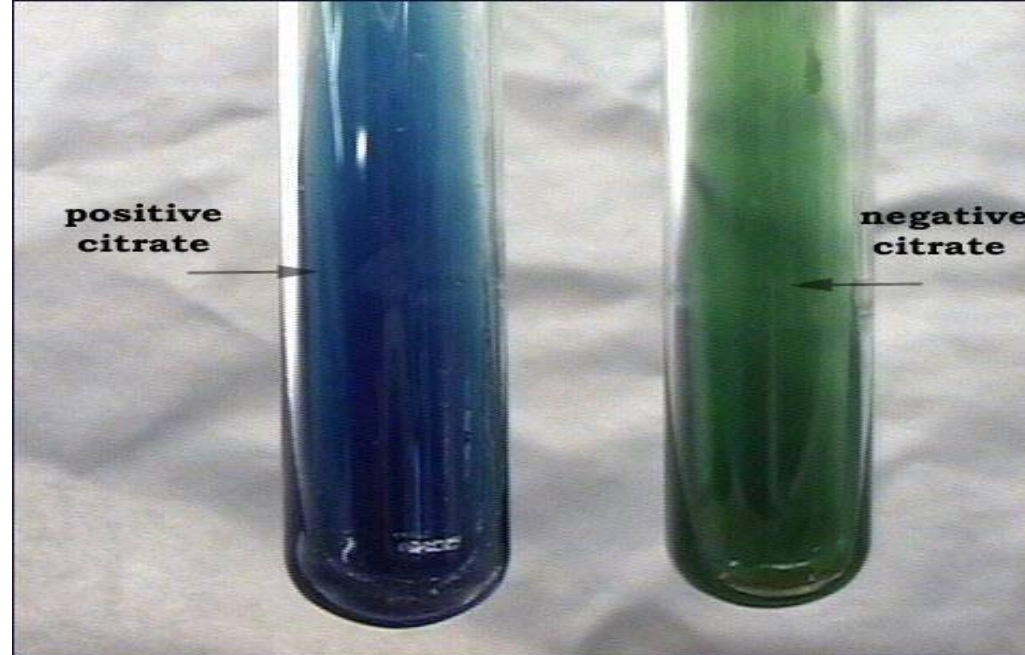


11–Simmon's Citrate Test:

Simmon's citrate agar is designed to assess whether a bacterium can utilize citrate as its sole carbon source and ammonium salts as its sole nitrogen source.

Organisms capable of using citrate produce alkaline by products that raise the pH of the medium, causing the indicator bromothymol blue to shift from **green** to an intense **blue**, signaling a **positive** citrate reaction.

This test is routinely used to differentiate among **Enterobacteriaceae** and other **Gram-negative rods**. Typical citrate-positive organisms include **Enterobacter**, **Klebsiella**, and some strains of **Proteus** and **Vibrio** (e.g., **Vibrio cholerae**).



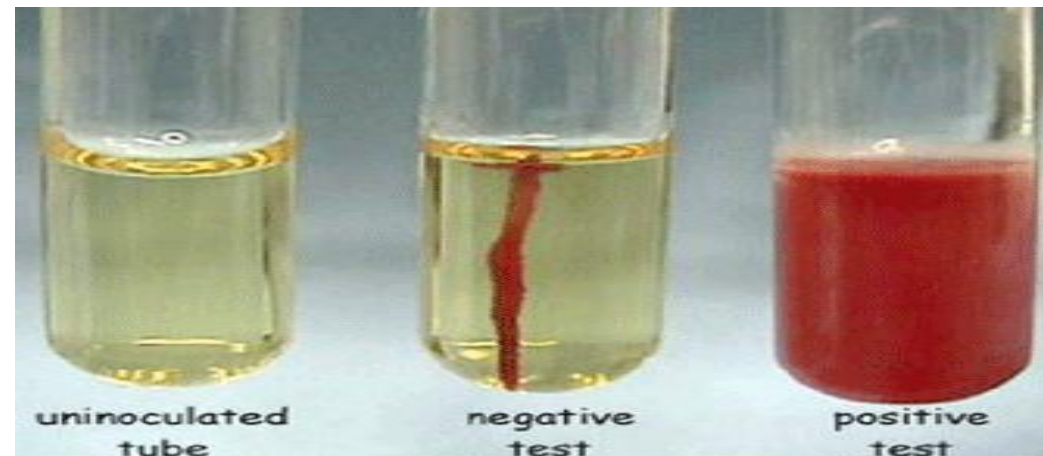
12–Motility Test (Motility Agar):

Motility agar is a **differential medium** used to determine whether a bacterium is **motile** by means of flagella.

The organism is inoculated with a straight stab into the medium.

- **Motile bacteria** will swim away from the stab line, causing the **entire medium to appear turbid**, indicating a positive result.
- **Non-motile bacteria** will only grow along the stab line, leaving the surrounding medium clear.

Common motile organisms include **Pseudomonas aeruginosa** and **Helicobacter pylori**.





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Any Question?



Thank you